



# Itron SDK for the Xcel Energy® 2030.5 Emulator

## New in This Document

| Revision | Date        | Description                                                             |
|----------|-------------|-------------------------------------------------------------------------|
| REV 001  | August 2022 | Added section on Audience; removed section "Using the Agent Simulator." |
| REV 000  | July 2022   | First date of publication                                               |

This document describes how to use the Itron Software Development Kit (SDK) for the Xcel Energy 2030.5 Emulator. The SDK provides sample source code and a test environment to assist software and hardware developers to develop IEEE 2030.5 Metering Function Set Clients to push real-time metering data to the cloud.

## Audience

This document is for software developers who are interested in building meter client applications to access real-time metering data.

## Overview

The Itron Gen™ 5 Riva™ meter is an Edge Computing platform with a state-of-the-art electricity sensor combined with a Wi-Fi radio and a 900 MHz or a 4G cellular radio.

The Edge Computing platform was designed to allow Itron and third-party software developers to run small software applications (agents) that allow the meter to locally process the raw metrology data to provide real-time insights and to control local energy assets such as solar inverters, batteries, and EV chargers. This data is stored on the cloud.

**Figure 1** Gen5 Riva meter

The Itron-provided smartphone client app provisions the meter to talk to the customer's Wi-Fi router and allow future 2030 Metering client devices, such as Amazon Echo dot or a Gateway on the local Wi-Fi network to successfully communicate with the meter and the 2030.5 HAN agent.

End-users must authenticate themselves to the utility for permission to get their Home Area Network (HAN) clients, such as smartphones or the Amazon Echo Dot, to communicate with the meter for real-time data via the utility portal.

The Itron 2030.5 HAN agent responds with the host name and usage point where metering data can be acquired. Local 2030.5 clients can then poll for data by sending an HTTP Get request.

## About the Itron Gen 5 Home Area Network (HAN)

Itron's Gen 5 HAN provides utility customers with real-time metering information to make better electricity consumption decisions by understanding the costs of appliances, heating, ventilation, and air conditioning (HVAC), and so on. The HAN also provides home and building automation systems with real-time feedback to minimize energy costs. The HAN is used to communicate with local energy assets such as solar photovoltaic systems, batteries, and EV chargers to monitor them and utilize the information to stabilize the grid.

The Gen 5 HAN, which is based upon the IEEE 2030.5 (SEP 2.0) specification, leverages a meter's Wi-Fi radio and the customer's Wi-Fi router to allow local 2030.5 clients to communicate with the meter. Itron has developed an agent, the 2030.5 HAN agent, that provides an IEEE 2030.5 metering function set server to enable 2030.5 metering clients to acquire real-time metering data over Wi-Fi.

The subset of the metering function set attributes supported in this release are as follows:

- **Instantaneous Demand** – How many kilo watts (kWh) of electricity is the meter measuring currently. This value is a signed value. If the meter detects more electricity being generated at the location being metered than being consumed, the value will be negative. If the value is positive, this represents the amount of power being delivered by the utility to the location.

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- **Current Summation Delivered** – The total kWh delivered by the utility to the location being measured since the meter was manufactured.
- **Current Summation Received** – The total kWh generated at the location being metered and returned to the power grid since the meter was manufactured.

## HAN Security

IEEE 2030.5 security is administered through message encryption and an Access Control List (ACL).

The IEEE 2030.5 specification states the following:

“Securing transactions between clients and servers is based on using HTTP over TLS (aka HTTPS) using TLS version 1.2. The TLS records are then transported using TCP. The TLS handshake mechanism provides mutual authentication based on device certificates or self-signed certificates and TLS records provide encryption and message authentication using the AES-CCM mode of operation. Access control lists allow or deny use of resources based on authentication level and address information. A registration list is used for authorizing clients.”

The security components are as follows:

- **Transport Layer Security (TLS) 1.2** – an Internet cryptographic protocol that utilizes both public and private key cryptography to provide privacy and data integrity. It is utilized by all websites that do any sort of commerce or the company that fails to use it may receive steep fines as it is a secure protocol that protects the data going between a client and a server. TLS 1.2 is the latest version currently accepted worldwide. TLS 1.2 utilizes the 2030.5 certificates for the client and the server for authentication and encryption using HTTP over TLS.
- **2030.5 certificate** - A digital certificate installed within a device that binds the device identity to the device. Device Certificates are exchanged by network access control and application protocols to authenticate devices as genuine IEEE 2030.5 and further to prove specific device identity. 2030.5 encryption key used by TLS - TLS\_ECDHE\_ECDSA\_WITH\_AES\_128\_CCM\_8.
- **Self-signed certificate** – clients used in the system can use 2030.5 certifications that were generated by the client. Self-signed certificates are temporary security certificates with a life of 3 years, after which they must be discarded and a new one created to take its place.
- **LFDI** – The Long Form Device Identifier is a SHA256 hash of the device’s certificate. It is a unique 20-byte value that is used by the Itron 2030.5 HAN agent to identify authorized 2030.5 Metering clients. The client must generate its own LFDI and communicate it to

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the Itron 2030.5 HAN Agent out-of-band to be placed in the agent's ACL.

- **ACL**– The ACL is a 'Whitelist' of client LFDIs that are authorized to communicate with the Itron 2030.5 HAN agent. If a 2030.5 client attempts to use mDNS to discover the 2030.5 HAN Agent's Metering server and its LFDI is not in the agent's ACL, the request is ignored. Should a client that is not in the ACL use the IP address, port, and usage point of the 2030.5 HAN agent and attempt to request metering data, the request will be ignored.
- **Port Usage** – mDNS discovery is done by sending a discovery request to port 5353 of the meter's IP address. RESTful commands (GET, PUT, POST, DELETE) must be made to port 8081 and to the proper usage point that is discovered by the client during mDNS discovery.
- **Logging and alerting** (logon/logoff, and other information that might indicate an attempt to compromise.

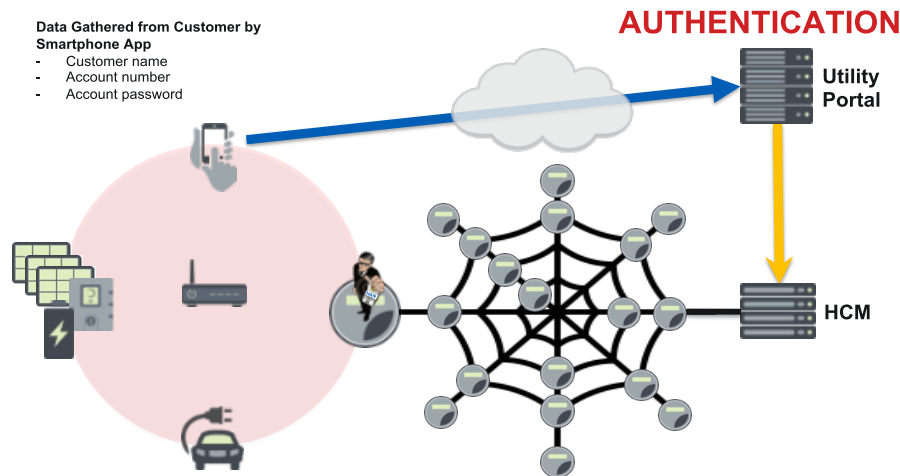
## Utility Portal

The utility portal is the customer's interface to the utility meters. The utility portal, which is a backend server authenticates clients for the meter, creates a unique identifier called LFDI . It gathers the Wi-Fi router name (SSDI) and password from the user and forwards them to the utility portal. The portal then sends the information to HCM to forward to the meter. The portal performs the following key services:

### Authentication

- Authenticates utility customers, matches utility customers with their meters, and gathers customer Wi-Fi and client ID information
- Provisions the customer's meter joining the customer's Wi-Fi network
- Deprovisions the customer's meter from the customer's Wi-Fi network
- Registers the 2030.5 clients to the meter's ACL
- Deregisters the 2030.5 clients to the meter's ACL

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The smartphone app gathers sufficient customer data to authenticate the user to the utility portal as a valid utility customer. It provides that data to the utility portal, requests authentication, and provides key information for meter provisioning to the Wi-Fi network and for registration of one or more 2030.5 clients to the meter/2030.4 HAN agent.

**Note:** Non-utility 2030.5 metering clients can only communicate with the meter/HAN agent once they are registered to the meter's ACL with the help of the utility smartphone app. The utility smartphone app performs all of the major functions: customer authentication, meter provisioning to the customer Wi-Fi router, deprovisioning, client registration and deregistration.

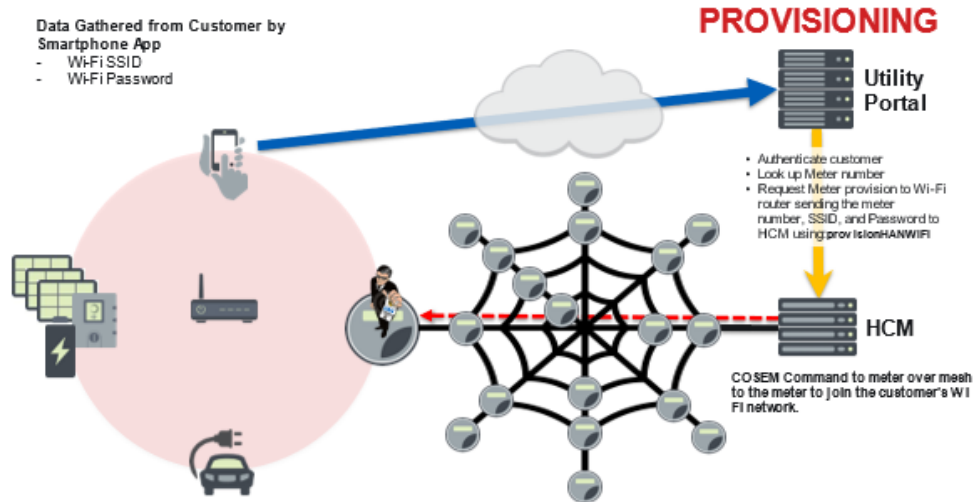
The utility smartphone app provides a means of entering the 2030.5 client's LFDI to be registered to the meter so that the new client may start its communication with the meter/2030.5 HAN agent.

## Provisioning

Provisioning is the process of providing the meter with the Wi-Fi credentials to enable it to join the customer's Wi-Fi network and be accessible to other devices on the local Wi-Fi network. The utility portal takes the Wi-Fi credentials and passes it along with the meter number to HCM. HCM then sends that information over the utility's private wireless network to the meter and invokes the meter command to join the Wi-Fi network.

The smartphone application initiates the provisioning process by collecting and sending the Wi-Fi Service Set Identifier (SSI) and password to the utility portal. The portal then looks up the meter number and sends down the provisioning request to the meter so that it will then attempt to join the requested Wi-Fi network.

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**Figure 2** HAN provisioning

If the meter is successful in joining the Wi-Fi network, it returns a 'Success' code to the utility portal. If the meter is unsuccessful in joining the Wi-Fi network, it returns a 'Failed to Join' code to the utility portal.

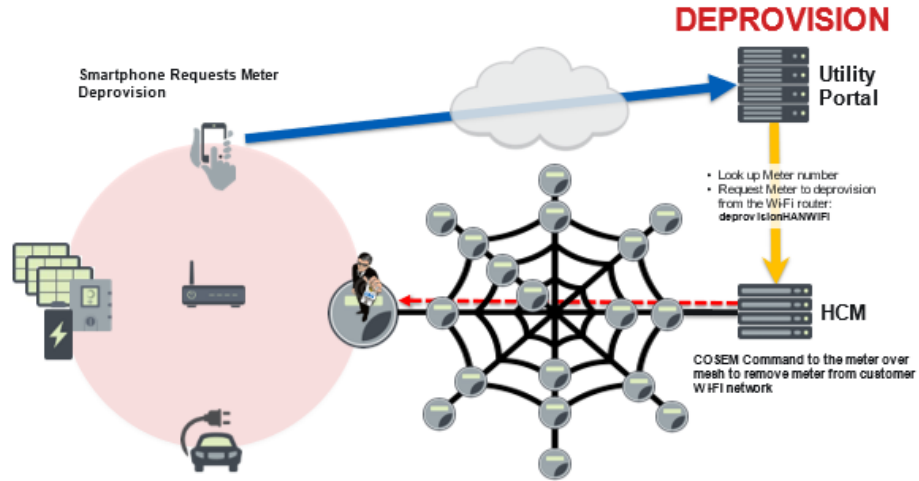
Reasons for failure include:

- The meter is too far away from the Wi-Fi router to join the network
- The Wi-Fi router password was incorrectly entered by the user
- Too much 2.4GHz interference
- The meter can only join a 2.4GHz Wi-Fi network – it will not talk to Wi-Fi devices over a 5.0GHz Wi-Fi network.

## Deprovisioning

Deprovisioning is the process of getting the meter to leave the customer's Wi-Fi network. The customer or the utility can start the deprovisioning process either from the smartphone application or from the utility portal admin. The portal sends a deprovisioning command to the customer's meter and the meter will remove itself from the customer's Wi-Fi network.

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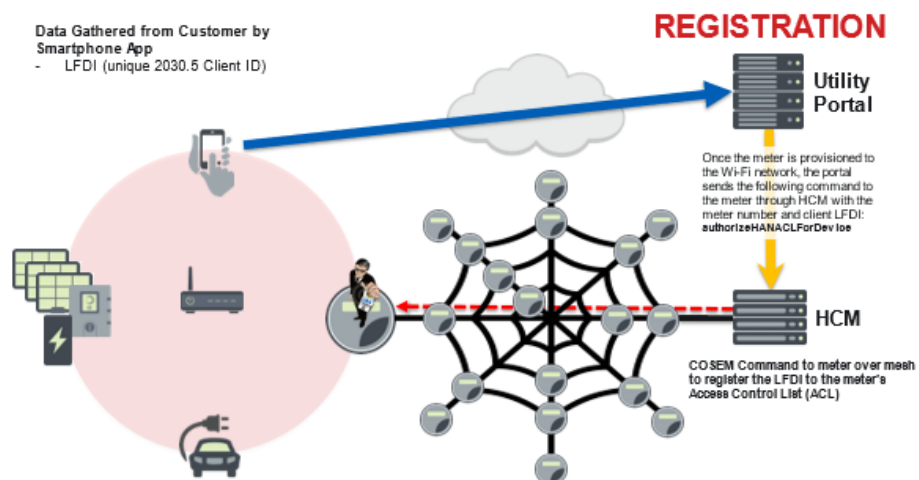
**Figure 3** Deprovisioning

## Registration

Metering client registration is the process of entering the client's unique identification number into the meter's ACL to enable the client to talk with the HAN agent.

IEEE 2030.5 clients cannot automatically talk to the 2030.5 HAN agent once the meter is on the customer's Wi-Fi network. The 2030.5 client's LFDI must be added to the meter's ACL, or else all mDNS and HTTP requests from the client(s) to the 2030.5 HAN agent/meter will be ignored.

The smartphone application initiates this process by providing its LFDI (or additional 2030.5 clients) to the utility portal and requests that they be registered to the meter.

**Figure 4** Metering client registration

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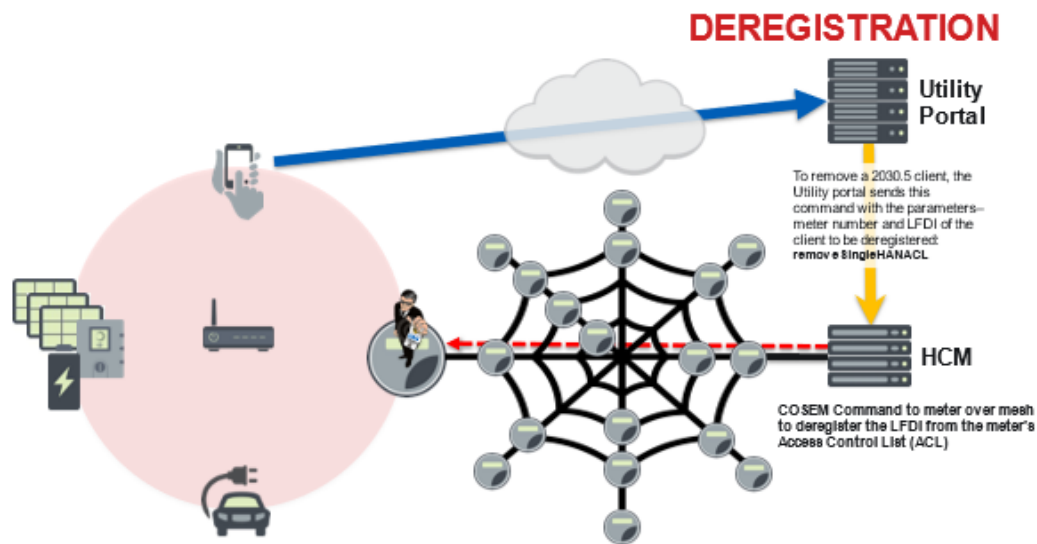
The LFDIs are sent to the meter out-of-band, just like the Wi-Fi SSID/password, so that the ACL is generated and securely maintained.

Once the client's LFDI is in the meter's ACL, the client can discover the metering server and begin requesting metering data.

## Deregistration

Deregistration is the process of removing one or more client LFDIs from the meter's ACL. This can be initiated by the customer or the utility.

**Figure 5** Deregistration



## Metering Server Discovery

The previous sections were an overview of how the non-2030.5 aspects of HAN work. As a developer of a new 2030.5 Metering client, your client must:

1. Include or self-sign a 2030.5 certificate
2. Create an LFDI and present it in a QRCode and as human readable 40 character alphanumeric sequence.
3. Perform multicast DNS (mDNS) discovery of the 2030.5 Metering server.
4. Use HTTP RESTful API to properly request metering data

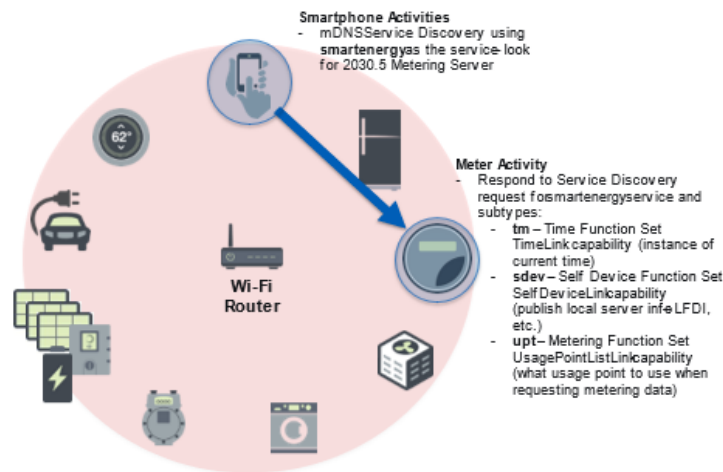
The 2030.5 HAN agent uses the mDNS protocol to discover the metering server. mDNS is the same service used by Apple, Hewlett Packard, and other manufacturers to discover wireless printers and other local Wi-Fi devices.

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The 2030.5 client discovers the metering server using the mDNS by sending a multicast message “\_smartenergy.\_tcp.local.” to port 5353. The 2030.5 metering server (2030.5 HAN agent) will respond with the hostname and usage point to which future metering data can be acquired.

**Figure 6** Metering server discovery



## Polling for Data

The 2030.5 client can start polling for data by sending the following HTTP 'Get' request to port 8081:

### Client Request

```
GET /upt/0/mr?s=0&l=10
// meter reading, up to 10 items
Host: {hostname}
Accept: application/sep+xml
```

The 2030.5 HAN Agent or Metering server will respond similarly as follows:

### Server Response

```
HTTP/1.1 HTTP/1.1 200 OK // acknowledgement of HTTP
Content-Type: application/sep+xml
Content-Length: {contentLength}

<MeterReadingList all="1" href="/upt/0/mr" results="1" subscribable="0"
xmlns="urn:ieee:std:2030.5:ns"> // meter reading results, only 1 item
  <MeterReading href="/upt/0/mr/0">
    <mRID>1500006CC8</mRID> // Instantaneous Demand read value
    <description>Instantaneous Reading for Wh</description>
    <ReadingLink href="/upt/0/mr/0/r"/> // first read item
```

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```

        <ReadingTypeLink href="/upt/0/mr/0/rt"/>
    </MeterReading>
</MeterReadingList>

```

The smartphone app source code has an working example of how to poll for and read values for Instantaneous Demand, Current Summation Delivered and Current Summation Received.

For more detail on the 2030.5 Metering function set server requests and responses, refer to the IEEE 2030.5 specification, Annex C, section C.10: Metering: Reading.

## IEEE 2030.5 Protocol

IEEE 2030.5, which is an application layer protocol that runs on top of a TCP/IP. 2030.5 uses the existing TCP/IP services, such as TCP, UDP, and TLS to pass information between the 2030.5 client(s) and the 2030.5 Metering server (2030.5 HAN agent).

| Layer            | Function                                                                 | Example                   |                            |
|------------------|--------------------------------------------------------------------------|---------------------------|----------------------------|
| Application (7)  | Services that are used with end-user applications                        | SMTP                      | IEEE 2030.5                |
| Presentation (6) | Formats the data so it can be viewed by the user.<br>Encrypt and Decrypt | JPG, GIF, HTTP, SSL, TLS  | TCP/IP                     |
| Session (5)      | Establishes and ends connections between two hosts                       | NetBIOS, PPTP             | TCP/IP                     |
| Transport (4)    | Responsbile for the transport protocol and error handling                | TCP, UDP                  | TCP/IP                     |
| Network (3)      | Reads the IP address from the packet                                     | Routers, Layer 3 switches | TCP/IP                     |
| Data Link (2)    | Reads the MAC address from the data packet                               | Switches                  | IEEE 802.11 - Mac          |
| Physical (1)     | Send data on to the physical wire                                        | Hubs, NICs, Cable         | IEEE 802.11– 2.4 GHz Radio |

IEEE 2030.5 defines the data schema or data structure organization and methods for reading and writing to/from the individual function set (Metering, Billing, Messaging, etc.) data structures.

2030.5 also defines the security tools and techniques to protect the data and the devices.

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# Glossary

## A

### ACL

Access Control List – list of approved HAN devices that exists in the meter

## C

### COSEM

Companion Specification for Energy Metering – transport and application layers of DLMS protocol

### Current summation delivered

Summation of kWh of electricity that have been delivered and consumed by the ratepayer since meter initialization.

### Current Summation Received

Summation of kWh of electricity that have been given back to the grid by the ratepayer's local generation since meter initialization.

## F

### Function Set

Logical grouping of resources that cooperate to implement 2030.5 features

## H

### HCM

HAN Communications Manager – the backend programmatic interface to the meters to manage HAN devices

## I

### Instantaneous Demand

kWh of power being demanded right now.

## L

### LFDI

Long Form Device Identifier – 20-byte hash of device's 2030.5 certificate used for HAN device access control

## M

### mDNS

Multicast DNS that resolves hostnames to IP addresses within small networks that do not include a local name server; standard method of finding available network services.

## P

### PSK

Pre-Shared Key – Wi-Fi network password

## S

### Self-Signed Certificate

A security certificate that is not signed by a certificate authority (used in smartphone apps)

### SSID

Service Set ID – name of a Wi-Fi network

### SunSpec Profiles

Industry group definitions as to how the function sets will be utilized by both the clients and the servers and the mutually agreed-upon tests that will demonstrate their compliance.

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## U

### **Utility Portal**

Xcel's backend interface to add HAN devices to your network.

## X

### **xmDNS**

eXtended mDNS extends the mDNS specification to site-local scope in order to support multi-hop LANs that forward multicast packets, but do not provide unicast DNS service.